MA1522 Linear Algebra for Computing
L2 Lesson 11



Tan Kai Meng

Semester 1, 2026/27

Recall that:

Let V be a subspace of $\mathbb{R}^n$. A *basis* S for V is a subset of V such that

- $\text{span}(S) = V$;
- S is linearly independent.

Informally, a basis for V is a collection of representatives of V such that

(a) every element of V is a linear combination of the vectors in the basis;

(b) having as few elements as possible subject to (a).

Note

There are usually many possible bases for V.

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Let V be a subspace of $\mathbb{R}^n$. The **dimension of** V, denoted $\dim(V)$, is the number of elements in a basis for V.

Theorem

Let V be a subspace of $\mathbb{R}^n$, and let $S \subseteq V$.

1. *If S is linearly independent, then there exists a basis T for V such that $S \subseteq T$. In particular,*
$$|S| \leq |T| = \dim(V).$$

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Corollary

Let V be a subspace of $\mathbb{R}^n$, and let $S \subseteq V$.

1. *If $|S| < \dim(V)$, then $\text{span}(S) \neq V$.*
2. *If $|S| > \dim(V)$, then S is not linearly independent.*
3. *If $|S| = \dim(V)$, then S is linearly independent if and only if $\text{span}(S) = V$.*

Corollary

Let V be a subspace of $\mathbb{R}^n$, and let $S \subseteq V$. Then S is a basis for V if any two of the following three statements hold:

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Let V and W be subspaces of $\mathbb{R}^n$. Which of the following statements is/are true:

a. If $V \subseteq W$, then $\dim(V) \leq \dim(W)$.

b. If $\dim(V) \leq \dim(W)$, then $V \subseteq W$.

c. If S is a basis for V and T is a basis for W, then
$$\dim(\text{span}(S \cup T)) \leq \dim(V) + \dim(W).$$

d. If S is a basis for V, T is a basis for W, and $S \cap T = \emptyset$, then
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a. If $V \subseteq W$, then $\dim(V) \leq \dim(W)$.

Let S be a basis for V. Then S is linearly independent and $S \subseteq W$, so there exists a basis T for W such that $S \subseteq T$. Thus

$$\dim(V) = |S| \leq |T| = \dim(W).$$

b. If $\dim(V) \leq \dim(W)$, then $V \subseteq W$.

Let $V = \text{span}(\{(1,0)\})$ and $W = \text{span}(\{(0,1)\})$. Then $\dim(V) = 1 = \dim(W)$ but $(1,0) \in V$ and $(1,0) \notin W$.

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$$\begin{aligned}\dim(\text{span}(S \cup T)) &\leq |S \cup T| = |S| + |T| - |S \cap T| \\ &\leq |S| + |T| = \dim(V) + \dim(W).\end{aligned}$$

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d. If S is a basis for V, T is a basis for W, and $S \cap T = \emptyset$, then $\dim(\text{span}(S \cup T)) = \dim(V) + \dim(W)$.

Let $V = \mathbb{R}^2$ with basis $S = \{(1,0),(0,1)\}$, and $W = \text{span}\{(1,1)\}$ with basis $T = \{(1,1)\}$. Then $S \cap T = \emptyset$ and $\text{span}(S \cup T) = \mathbb{R}^2$. Thus

$$\dim(\text{span}(S \cup T)) = \dim(\mathbb{R}^2) = 2,$$
$$\dim(V) + \dim(W) = 2 + 1 = 3.$$

Coordinates Relative To A Basis

Let V be a subspace of $\mathbb{R}^n$, with basis $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$.

For each $\boldsymbol{v} \in V$, there exist unique $c_1, c_2, \ldots, c_k$ such that

$$\boldsymbol{v} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k.$$

Let $[\boldsymbol{v}]_S := \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix}$. We call $[\boldsymbol{v}]_S$ the **coordinates of $\boldsymbol{v}$ relative to** S.

Then $\boldsymbol{v} = \boldsymbol{A}[\boldsymbol{v}]_S$, where $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k \end{pmatrix}$.

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Question 2

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Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ and $T = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, where

$$\begin{aligned} \boldsymbol{u}_1 &= (1, 2, -1), \quad \boldsymbol{u}_2 = (0, 2, 1), \quad \boldsymbol{u}_3 = (0, -1, 3), \\ \boldsymbol{v}_1 &= (1, 5, 4), \quad \boldsymbol{v}_2 = (-1, 3, 7), \quad \boldsymbol{v}_3 = (2, 2, 4). \end{aligned}$$

a. Prove that both S and T are bases for $\mathbb{R}^3$. (Hint: Pg 108 of Chpt 3.)

b. Let $\boldsymbol{w} = (1, 1, 1)$. Compute $[\boldsymbol{w}]_S$ and $[\boldsymbol{w}]_T$. (Hint: Pg 118–9 of Chpt 3.)

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   Check that $\det \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{pmatrix}, \det \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{pmatrix} \neq 0.$

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   $\operatorname{RREF} \left(\begin{array}{ccc|c} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 & \boldsymbol{w} \end{array} \right) = \left(\begin{array}{c|c} \boldsymbol{I}_3 & \begin{matrix} 1 \\ -\frac{1}{7} \\ \frac{5}{7} \end{matrix} \end{array} \right)$. So $[\boldsymbol{w}]_S = \begin{pmatrix} 1 \\ -\frac{1}{7} \\ \frac{5}{7} \end{pmatrix}$.

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Transition Matrix

Let V be subspace of $\mathbb{R}^n$, and let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ and $T = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$ be two different bases for V.
The **transition matrix from S to T** is the matrix

$$\boldsymbol{P} := \begin{pmatrix} [\boldsymbol{u}_1]_T & [\boldsymbol{u}_2]_T & \cdots & [\boldsymbol{u}_k]_T \end{pmatrix}.$$

Theorem

For all $\boldsymbol{w} \in V$, we have

$$[\boldsymbol{w}]_T = \boldsymbol{P}[\boldsymbol{w}]_S.$$

Question 3

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Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ and $T = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, where

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You may assume that both S and T are bases for $\mathbb{R}^3$.

a. Compute the transition matrix from S to T.

b. Let $\boldsymbol{w} \in \mathbb{R}^3$ and suppose that $[\boldsymbol{w}]_S = \begin{pmatrix} 1 \\ -\frac{1}{7} \\ \frac{5}{7} \end{pmatrix}$. Use part (a) to compute $[\boldsymbol{w}]_T$.

(Hint: Pg 150–1, 148 of Chapter 3.)

Question 3: Solution

a. The transition matrix $\boldsymbol{P}$ from S to T equals $\begin{pmatrix}[\boldsymbol{u}_1]_T & [\boldsymbol{u}_2]_T & [\boldsymbol{u}_3]_T\end{pmatrix}$.

Each $[\boldsymbol{u}_i]_T$ satisfies $\begin{pmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3\end{pmatrix}[\boldsymbol{u}_i]_T = \boldsymbol{u}_i$.

$$\operatorname{RREF}\left(\begin{array}{ccc|c|c|c}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3\end{array}\right) = \left(\begin{array}{c|c|c|c}\boldsymbol{I}_3 & \begin{matrix}\frac{3}{4}\\ -\frac{1}{2}\\ -\frac{1}{8}\end{matrix} & \begin{matrix}\frac{1}{2}\\ 0\\ -\frac{1}{4}\end{matrix} & \begin{matrix}-\frac{3}{4}\\ \frac{1}{2}\\ \frac{5}{8}\end{matrix}\end{array}\right)$$

Thus $\boldsymbol{P} = \begin{pmatrix}\frac{3}{4} & \frac{1}{2} & -\frac{3}{4}\\ -\frac{1}{2} & 0 & \frac{1}{2}\\ -\frac{1}{8} & -\frac{1}{4} & \frac{5}{8}\end{pmatrix}$.

b. $[\boldsymbol{w}]_T = \boldsymbol{P}[\boldsymbol{w}]_S = \begin{pmatrix}\frac{3}{4} & \frac{1}{2} & -\frac{3}{4}\\ -\frac{1}{2} & 0 & \frac{1}{2}\\ -\frac{1}{8} & -\frac{1}{4} & \frac{5}{8}\end{pmatrix}\begin{pmatrix}1\\ -\frac{1}{7}\\ \frac{5}{7}\end{pmatrix} = \begin{pmatrix}\frac{1}{7}\\ -\frac{1}{7}\\ \frac{5}{14}\end{pmatrix}$.

Question 3: Solution

a. The transition matrix $\boldsymbol{P}$ from S to T equals $\begin{pmatrix}[\boldsymbol{u}_1]_T & [\boldsymbol{u}_2]_T & [\boldsymbol{u}_3]_T\end{pmatrix}$. Each $[\boldsymbol{u}_i]_T$ satisfies $\begin{pmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3\end{pmatrix}[\boldsymbol{u}_i]_T = \boldsymbol{u}_i$.

$$\operatorname{RREF}\left(\begin{array}{ccc|c|c|c} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{array}\right) = \left(\begin{array}{c|c|c|c} \boldsymbol{I}_3 & \begin{matrix}\frac{3}{4}\\ -\frac{1}{2}\\ -\frac{1}{8}\end{matrix} & \begin{matrix}\frac{1}{2}\\ 0\\ -\frac{1}{4}\end{matrix} & \begin{matrix}-\frac{3}{4}\\ \frac{1}{2}\\ \frac{5}{8}\end{matrix} \end{array}\right)$$

Thus $\boldsymbol{P} = \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & -\frac{3}{4} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} & \frac{5}{8} \end{pmatrix}$.

b. $[\boldsymbol{w}]_T = \boldsymbol{P}[\boldsymbol{w}]_S = \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & -\frac{3}{4} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{8} & -\frac{1}{4} & \frac{5}{8} \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{1}{7} \\ \frac{5}{7} \end{pmatrix} = \begin{pmatrix} \frac{1}{7} \\ -\frac{1}{7} \\ \frac{5}{14} \end{pmatrix}$.

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Next Lesson ...

We will explore **Sections 4.1 and 4.2**.
Please watch/review the video clips for these sections before then.